Faculty of Computing

Year 1 Semester 2 (2025)

SE1020 – Object Oriented Programming Lab Sheet 05

Question 1

1. Develop a system to manage details for a Pizza Restaurant System.

   Scenario:

   The system should record pizza order details, calculate the total price based on pizza size, handle optional toppings, and provide special features such as delivery charges and estimated preparation time.

   The pizza price chart is as follows:

Size	**Price**
Small	$8.00
Medium	$10.00
Large	$12.00

   Each extra topping adds $1.50 to the base price.

   (a) Implement a class called "**PizzaOrder**" with the orderID (String), customerName (String), size (String), (Small, Medium, or Large), toppingCount (int), and isDelivery (boolean) as attributes.

   (b) Implement a **default constructor** that initializes attributes with default values.

   (c) Implement **overloaded constructors** to allow:

      i. a constructor that accepts only orderID and customerName.

      ii. A constructor that accepts all five attributes.

   (d) Provide **getters** and **setters** for size. The setter for size must validate that the size is either "Small", "Medium", or "Large". If an invalid size is entered, set the default to "Small".

   (e) Implement a method called getBasePrice() which return the price according to the size of the pizza (refer to the above tabledatails)

   (f) Implement **method overloading** by creating two versions of the *calculateTotalPrice()* method:

i. One that calculates the total price using the stored values, including toppings and delivery fee (if applicable). (Hint – call getBasePrice() method to get the basic price according to the size of the pizza.)

ii. Another that accepts a discountPercentage as parameter (double) and applies the discount to the total price. (Hint – call calculateTotalPrice() method to get the price according to the size, number of toppings, and delivery status.)

g) Implement a method called estimatePreparationTime() to calculate and return the preparation time based on the number of toppings.

h) Implement a method called displayOrderDetails() to display the full order details, including the total price and estimated preparation time.

[Hint: Use the sample main method given below to implement scenario]

Sample main method:

```java
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    // First order
    PizzaOrder order1 = new PizzaOrder("03001", "Olivia Green", "Medium", 3, true);
    order1.displayOrderDetails();

    // Second order with discount
    PizzaOrder order2 = new PizzaOrder("03002", "Noah Smith", "Large", 2, false);
    order2.displayOrderDetails();
    System.out.println("Enter the discount percentage: ");
    double percentage = sc.nextDouble();
    System.out.println("Order 02 - Price after the discount: $" + order2.calculateTotalPrice(percentage));
    System.out.println();

    // Third order using default constructor
    PizzaOrder order3 = new PizzaOrder();
    order3.displayOrderDetails();
}
```

Question 02

A new Cable TV company called VTV is providing over 100 entertainment TV channels to customers. Any person can become a customer by registering with VTV. The customer has to provide customer details and the package (channels he wishes to see) during the registration. The customer is provided with a unique customer number. The customer details are stored in the customer table and the package details selected by the customer are stored in the subscription table.

Once a month the VTV sends the customer a detailed bill. This is produced using the data stored in the Subscription Table, Customer Table, and Channel details Table. The customer can login to VTV's website and perform several online tasks. The customer can add new channels to the subscription he has. He can remove channels from his subscription. In both cases the Subscription Table should be updated. In addition the customer can provide a list of channels that he wants to get a detailed weekly TV schedule by email. These details are also stored in the Subscription Table.

The customer can pay the monthly bills online using a credit card. He has to provide the credit card details and the amount to the system. The payment details are stored in

the Payments Table. Every week VTV sends customers who have requested detailed TV schedules the details by email. TV schedules are stored in the Channel details Table.

Do the Noun Verb analysis to identify the classes in the description.

a) Identify all the Nouns which can be potential classes

b) Identify the set of potential classes, show clearly how the nouns were rejected using the five rules.

c) Draw the CRC cards for any 5 classes you have identified in part (b)

Question 03

Please submit your answer to the question below to Git. When committing the file, ensure that you include a meaningful commit message (e.g., "Lab05 - Question03").

An Online bookshop sells different things. Currently it has books, magazines and CDs. All these products have some common types of information such as price, title etc.. Also each product may have specific information such as author, number of pages, artists, type etc..) .

The registered customers can place orders for products. The ordered products are usually delivered together, but sometimes several shipments are needed when the store runs out of products for a while. An order may contain several products.

The products are delivered to personal customers and payments are done at the delivery. The business customers will get their orders directly, and they will be sent an invoice afterwards. Both customer types can make the payment also by using credit card. When registering, a customer is prompted to give contact and personal information needed for authentication, delivery of orders and invoicing.

The bookshop system must be able to add and remove a customer, add new product and view/update existing product, make or remove an order and view existing order, view and generate reports on the products.

Do the Noun Verb analysis to identify the classes in the description.

(a) Identify all the Nouns which can be potential classes

(b) Identify the set of potential classes, show clearly how the nouns were rejected using the five rules.